Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/03/2017
PI Name:	Hogan, Harry Ph.D.		
Project Title:	Pre-Treatment Approaches for Improving the Response of Bone to Disuse and Recovery		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	77843-3123	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2018 per NSSC information (Ed., 8/22/17) NOTE: End date changed to 8/31/2017 per NSSC information (Ed., 6/3/16) NOTE: Period of performance dates corrected per NSSC information (Ed., 9/10/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Allen, Matthew Ph.D. (Indiana University) Bloomfield, Susan Ph.D. (Texas A&M University) Fluckey, James Ph.D. (Texas A&M University)		
Grant/Contract No.:	NNX13AM43G		
Performance Goal No.:			
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Task Description:	This research proposal is aimed at better understanding the basic bone biological effects of various osteoporosis treatments as therapies for reducing or preventing bone loss due to space flight. The project will study hindlimb unloaded (HU) rats, which is a technique that is well established as an effective and informative ground-based analog for studying the effects of space flight on the musculoskeletal system. In order to better understand the detailed effects of each, the treatments will be studied separately. The four treatments to be examined include two anabolic (bone building) therapies and two anti-resorptive therapies. One anabolic approach will use voluntary exercise and the other a new osteoporosis drug (anti-sclerostin antibody). Anti-resorptive drugs are the most popular for treating clinical osteoporosis and achieve their benefits by preventing bone resorption, which is part of the normal cycle of bone replenishment. Furthermore, the treatments will be used as pre-treatments and only administered prior to HU exposure. This is not realistic for astronauts, but it will allow more in-depth study of the fundamental nature of each treatment, and it will also align with the anticipated requirements of possible follow-up space flight experiments using the same rat animal model. The experiments to be conducted will characterize beneficial effects of pre-treatment upon bone loss due to HU as well as bone recovery following HU and return to weightbearing. Bone properties will be quantified using CT-based analyses, histological techniques for assessing bone cell function, as well as biomechanical testing to measure strength.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Completion of the studies outlined in this research proposal will provide new insights into the basic nature of the effects of exercise and pharmaceutical agents and how these interact with mechanical unloading and recovery therefrom. Removal of weightbearing through hindlimb unloading is a severe and acute form of mechanical disuse that has been shown to simulate many of the effects of microgravity exposure due to space flight on the musculoskeletal system. Hindlimb unloading also induces many musculoskeletal system changes that are similar to accelerated aging as well as other disuse scenarios such as stroke, spinal cord injury, or neuromuscular impairment. Thus, insights gained should also help advance basic understanding of the fundamental biology associated with Earth-bound clinical medicine as well as space flight effects. Future space flight experiments can be designed to replicate the research to be conducted in this proposed project, and comparing results from the two will provide essential and more detailed understanding of the full efficacy, as well as limitations, of the adult rat HU model as a ground-based analog for human space flight. Studying these treatments separately is not necessarily realistic for actual mission scenarios, but the results will nevertheless provide evidence-based rational data for contemplating and designing possible co-therapies using either combinations (simultaneously) or possible sequential scheduling protocols.
Task Progress:	The primary focus of effort during the current year has been on: (a) maintaining progress on the second set of animal experiments, and (b) continuing to conduct and complete analysis of data and interpretation of results. With reference to the experiments and specific aims as originally proposed (overviewed above), the order was reversed, and the anti-resorptive pre-treatment portion of the study was done first. The animal work for the anabolic pre-treatment portion of the project was originally scheduled to start in Spring of 2015, but this was delayed significantly due to acquisition of the sclerostin anti-body agent and establishment of proper dosing. Considerable time and effort were expended during year 2 on this task, and a Material Transfer Agreement (MTA) was finally executed with Lilly near the very end of Year 2 (on July 24, 2015). Accordingly, the focus of animal work during Year 3 and the current year has been on the anabolic pre-treatment part of the project. Animal groups and generic timelines are summarized in the schematic below. Anabolic Pre-Treatment Animal Study: Four experimental groups are specified for each of the two sets of animal experiments. The two anabolic pre-treatments are pharmaceutical and mechanical. Specifically, the pharmaceutical agent used was an anti-sclerostin anti-body compound provided by Lilly (Eli Lilly and Company, Indianapolis, IN) through a Material Transfer Agreement. The mechanical anabolic consists of a voluntary rat jumping protocol intended to simulate resistance exercise. The four groups for this part of the study are: Ambulating Control (CON) ; HU Control (HUC) ; Sclerostin Anti-body (SAB) ; Exercise (REX) In vivo measures were made on all animals every 28 days. These consist primarily of pQCT (peripheral Quantitative Computed Tomography) scans conducted on anesthetized animals at the proximal tibia metaphysis (PTM) and at the tibia diaphysis (TD), or midshaft. In addition, blood serum samples have been acquired at each 28-day interval and stored for possible future analysis of bone activity biomarkers. A host of additional analyses are conducted ex vivo on bone specimens harvested from the animals following euthanasia at each of the time 28-day time points. Ex Vivo Endpoint Measures: Densitometry and microstructure by microCT (tibia and femur; metaphyses, midshaft, FN); ~ trabecular microarchitecture, vBMD, BMC, TMD, etc. Static and dynamic histomorphometry (tibia and femur; metaphyses, midshaft, FN); ~ trabecular, cortical, MS/BS, MAR, BFR, etc. Bone Strength (tibia metaphysis RPC, midshaft; femur midshaft, FN); ~ ultimate load, stiffness, ultimate strength, elastic modulus, toughness, etc. Muscle Weights (hindlimb posterior crural muscles); ~ wet weights for gastrocnemius, soleus, and plantaris Major Milestones: Milestones are shown below summarizing the tasks and timing of effort for the current year, with some overlap going back to the latter portion of Year 3 for context and continuity. Jan. – Sep. 2016: • Animal work for Scl-Ab drug pre-treatment conducted in 3 cohorts: Cohort 1 – 26 Jan 2016 to 22 June 2016 (20 Scl-Ab; 5 REX) ; Cohort 2 – 18 Apr 2016 to 31 Aug 2016 (13 Scl-Ab) ; Cohort 3 – 5 May 2016 to 14 Sept 2016 (12 Scl-Ab) Sep. – Dec. 2016: • Continuing refinement of voluntary jumping exercise details and new cages ; • Ex Vivo analyses (pQCT, microCT, mechanical testing, histomorphometry) Jan. – Sep. 2017: • Animal work for voluntary jumping exercise (REX) to be conducted in 4 cohorts: Cohort 1 – 10 Jan 2017 to 16 May 2017 (10 REX) ; Cohort 2 – 13 June 2017 to 14 Nov 2017 (10 REX) ; Cohort 3 – 13 Sept 2017 to 10 Feb 2018 (10 REX) ; Cohort 4 – 12 Dec 2017 to 8 Apr 2018 (10 REX) As reflected above, a decision was made to stagger the two parts of the anabolic pre-treatment arm of the project. The

	Scl-Ab (pharmacologic) pre-treatment part was done first and was completed in September 2016 (the start of the current year). The voluntary jumping exercise (REX) pre-treatment part was resumed in January 2017 and will continue into the next year (September 2017 to August 2018). A no-cost extension is being requested to maintain spending capabilities for ordering animals and housing costs in order to complete all of the needed animal experiments. All funds have already been received (as of April 2017). Two factors prompted the strategy to stagger the Scl-Ab and REX parts of the study. First, the duration of the window of availability for the sclerostin anti-body compound was uncertain, so it was deemed best to acquire all of the material that was promised as soon as possible and reasonable. The second factor is that the voluntary jumping exercise required additional refinements and adjustments in order to make it practical and effective for the remaining 40 animals scheduled to conduct the jumping exercises. Five animals were included in the first cohort of the Scl-Ab pre-treatment part, and the knowledge gained from that was used to adjust and improve the process. Most prominently, the customized cages required for acclimating, training, and exercising the animals needed to be redesigned and new ones constructed. Consequently, the first of the four remaining REX cohorts was not started until January 2017. Further, the labor-intensive nature of the jumping exercise training protocol dictates a maximum of 10 animals per cohort, as reflected in the summary above. The animal numbers shown above for Cohorts 1-4 of the exercise anabolic part are for the REX groups only. There will be at least 10 more animals per cohort in order to populate additional ambulating controls (CON) and hindlimb unloaded controls (HUC) groups. Thus, additional costs will be incurred beyond the current year end date of August 31, 2017, as follows: • Purchase of 40-45 more adult male rats for Cohorts 3 and 4 ; Animal housing for the remainder of Cohort 2 and all of Cohorts 3 and 4.
Bibliography Type:	Description: (Last Updated: 01/11/2021)
Abstracts for Journals and Proceedings	Lenfest SE, Kosniewski JL, Looper AL, Brezicha JE, Metzger CE, Black JM, Bloomfield SA, Hogan HA. "Initial Results for a New Positive Reinforcement Voluntary Jumping Exercise in Rats Show Enhanced Bone Parameters." Poster Presentation, Poster Session A: Bone, 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. Poster #17261. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017
Abstracts for Journals and Proceedings	Elizondo JP, Black JM, Brezicha JE, Lenfest SE, Kosniewski JL, Bloomfield SA, Allen MR, Hogan HA. "Effects of Bisphosphonate Pre-Treatments on Subsequent Unloading Response at the Tibia and Femur in Adult Rats." 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. Poster #17264. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017
Articles in Peer-reviewed Journals	Metzger CE, Brezicha JE, Elizondo JP, Anand Narayanan S, Hogan HA, Bloomfield SA. "Differential responses of mechanosensitive osteocyte proteins in forelimbs and hindlimbs in hindlimb unloaded rats." Bone. 2017 Aug 3. [Epub ahead of print] https://doi.org/10.1016/j.bone.2017.08.002 ; PubMed PMID: 28782619 , Aug-2017